Methylmalonic Aciduria and Homocystinuria, cblL Type (MAHCL) — Comprehensive Disease Report

Disease: Methylmalonic Aciduria and Homocystinuria, cblL Type **OMIM:** #620940 (MAHCL) · **MONDO:** MONDO:0975798 · **Gene:** *THAP11* (OMIM 609119; HGNC:20856; 16q22.1) **Category:** Mendelian · autosomal recessive inborn error of intracellular cobalamin (vitamin B12) metabolism

Summary

Methylmalonic aciduria and homocystinuria, cblL type (MAHCL), is an **ultra-rare autosomal-recessive inborn error of intracellular cobalamin (vitamin B12) metabolism** caused by biallelic hypomorphic variants in ***THAP11***, a THAP-domain–containing transcription factor located at chromosome 16q22.1. Unlike the classic combined methylmalonic aciduria/homocystinuria disorders that arise from defects in the cobalamin-processing enzymes and chaperones themselves (e.g., *MMACHC* in cblC), cblL is a **transcriptional/regulatory disorder**: THAP11 partners with the global co-regulator HCFC1 (and ZNF143) to activate transcription of *MMACHC*. When THAP11 function is impaired, *MMACHC* expression is downregulated, producing a **cblC-like biochemical phenotype (mild methylmalonic aciduria, elevated/high-normal homocysteine, low-normal methionine) despite an entirely normal *MMACHC* gene sequence.** This places cblL within the "cblX-like" group of cobalamin disorders alongside HCFC1-related cblX and ZNF143-related disease.

The disorder was defined by Quintana et al. (2017), who identified a homozygous missense variant, *THAP11* c.240C>G (p.Phe80Leu; F80L), in a boy of Moroccan parentage who presented at 2 months of age with **myoclonic seizures and severe global developmental impairment** after *MMACHC* and *HCFC1* mutations had been excluded [PMID: 28449119]. To date this remains the **only reported patient with cblL worldwide**, so the clinical spectrum, natural history, prognosis, and treatment response are largely **extrapolated from the biochemically equivalent cblC and cblX disorders.** Because THAP11 (also known as Ronin) is an essential developmental transcription factor — its complete loss causes peri-implantation lethality in

mice and craniofacial/neural defects in zebrafish — the human disorder is expected to be syndromic and neurodevelopmental, reflecting both the *MMACHC*-dependent cobalamin defect and *MMACHC*-independent disruption of THAP11's broad transcriptional program governing protein biosynthesis, energy metabolism, and neural/neural-crest development.

Rational management follows the cblC/cblX template: parenteral **hydroxocobalamin**, **betaine**, **folinic acid**, and **levocarnitine**, with dietary protein moderation. However, as in cblC, cognitive and neurological impairment frequently persists despite biochemical correction, and outcome data specific to cblL do not yet exist. This report compiles the current state of knowledge across all requested disease-characteristic domains, clearly distinguishing cblL-specific evidence (a single case plus model-organism data) from information imported from the wider cobalamin-disorder literature.

Key Findings

F001 — cblL (MAHCL, OMIM #620940) is caused by biallelic *THAP11* variants

cblL is genetically defined by **biallelic variants in *THAP11*** (16q22.1). The index and only case is a boy of Moroccan parentage carrying a homozygous missense variant, **c.240C>G (p.Phe80Leu; F80L)**, who presented at 2 months of age with myoclonic seizures and severe global developmental impairment; *MMACHC* (cblC) and *HCFC1* (cblX) mutations were specifically excluded before *THAP11* was sequenced [PMID: 28449119]. OMIM assigns cblL the number **#620940 (MAHCL)**, with the gene *THAP11* catalogued at **OMIM 609119**.

> "We sequenced THAP11 by Sanger sequencing and discovered a potentially pathogenic, homozygous variant, c.240C > G (p.Phe80Leu)." — Quintana et al. 2017 [PMID: 28449119]

In clinical practice, cblL sits within the combined MMA+homocystinuria (HC) gene panel. Hwang et al. (2021) list the relevant genes as *MMACHC* (cblC), *MMADHC* (cblD), *LMBRD1* (cblF), *ABCD4* (cblJ), *THAP11* and *ZNF143* (cblX-like), and *HCFC1* (cblX) [PMID: 34655177]:

> "biallelic variants in one of the following genes: MMACHC (cblC), MMADHC (cblD), LMBRD1 (cblF), ABCD4 (cblJ), THAP11 (cblX-like), and ZNF143 (cblX-like), or a hemizygous variant in HCFC1 (cblX)" — Hwang et al. 2021 [PMID: 34655177]

Ontology terms: MONDO:0975798 (cblL); gene HGNC:20856 (*THAP11*).

F002 — Mechanism: the THAP11–HCFC1 complex transcriptionally regulates *MMACHC*; its loss downregulates *MMACHC*

THAP11 is a **THAP-domain transcription factor** that partners with the global transcriptional co-regulator **HCFC1** to drive *MMACHC* expression. This is the pivotal mechanistic link that explains why a disorder with a normal *MMACHC* sequence nonetheless behaves biochemically like cblC. Quintana et al. (2017) state the axis directly:

> "HCFC1 regulates cobalamin metabolism via the regulation of MMACHC expression through its interaction with THAP11, a THAP domain-containing transcription factor." — Quintana et al. 2017 [PMID: 28449119]

The same regulatory logic was established in cblX, where *HCFC1* mutations produce a severe reduction in *MMACHC* mRNA and protein in patient fibroblasts:

> "The severe reduction in MMACHC mRNA and protein within subject fibroblast lines suggested a role for HCFC1 in transcriptional regulation of MMACHC" — Yu et al. 2013 [PMID: 24011988]

RNA-seq of the THAP11-mutant patient's fibroblasts showed **downregulation of *MMACHC* (and *TMOD2*)**, consistent with a cblC-like biochemical complementation pattern despite a normal *MMACHC* sequence. This makes cblL a **regulatory (upstream) phenocopy of cblC.**

Ontology terms: GO:0006355 (regulation of DNA-templated transcription); GO:0009235 (cobalamin metabolic process); GO:0003700 (DNA-binding transcription factor activity).

F003 — Downstream biochemistry: MMACHC is a cytosolic cobalamin chaperone performing decyanation/dealkylation

The functional consequence of reduced *MMACHC* is loss of the **cytosolic cobalamin trafficking chaperone** that prepares dietary/therapeutic cobalamin for use. MMACHC catalyzes **reductive decyanation** of cyanocobalamin and **glutathione-dependent dealkylation** of methyl- and adenosylcobalamin (turnover 11.7 h^{-1} for MeCbl and 0.174 h^{-1} for

AdoCbl at 20 °C), generating cob(I)alamin for downstream conversion to the two active cofactors — methylcobalamin (for methionine synthase, MTR) and adenosylcobalamin (for methylmalonyl-CoA mutase, MMUT) [PMID: 19801555].

> "MMACHC, a cytosolic cobalamin trafficking chaperone, has been shown recently to catalyze a reductive decyanation reaction when it encounters cyanocobalamin." — Kim et al. 2009 [PMID: 19801555]

When MMACHC activity falls, **both** downstream cofactors become deficient, impairing (a) methylmalonyl-CoA mutase → methylmalonic acid accumulation and (b) methionine synthase → homocysteine accumulation with low methionine. This is the biochemical signature of the "combined" MMA+HC disorders.

Ontology terms: CHEBI:17439 (cyanocobalamin), CHEBI:28115 (methylcobalamin), CHEBI:18408 (adenosylcobalamin), CHEBI:16009 (methylmalonic acid), CHEBI:17230 (homocysteine); GO:0005829 (cytosol).

F004 — Clinical phenotype and treatment framework (extrapolated from cblC)

The single reported cblL patient presented in **infancy (2 months)** with **myoclonic seizures and severe global developmental impairment**, with **mild methylmalonic aciduria and low-normal plasma methionine**; overt homocystinuria was not documented in this individual [PMID: 28449119]. Standard therapy for the biochemically equivalent cblC group comprises **hydroxocobalamin (OHCbl), betaine, folinic acid, levocarnitine, and dietary protein restriction** [PMID: 23746552]:

> "Hydroxocobalamin (OHCbl), betaine, folinic acid, levocarnitine and eventually dietary protein restriction are the main therapeutic approaches." — Matos et al. 2013 [PMID: 23746552]

In cblC, cognitive and visual impairment are nearly constant despite treatment, tempering expectations for cblL. The developmental/neural involvement underlying the severe infantile phenotype is supported by model data:

> "The loss of THAP11 in zebrafish embryos results in craniofacial abnormalities including the complete loss of Meckel's cartilage" — Quintana et al. 2017 [PMID: 28449119]

F005 — THAP11 (Ronin) is an essential HCF-1–partnered transcription factor for embryogenesis, pluripotency, and biosynthetic gene programs

THAP11 (Ronin) is a **THAP-domain DNA-binding factor essential for embryonic stem-cell self-renewal**. Ronin deficiency causes peri-implantation lethality and inner-cell-mass defects, and Ronin binds directly to HCF-1 [PMID: 18585351]:

> "its deficiency in mice produces periimplantational lethality and defects in the inner cell mass" — Dejosez et al. 2008 [PMID: 18585351]

Ronin/HCF-1 binds a **hyperconserved enhancer element** and upregulates genes for protein biosynthesis and energy production (transcription initiation, mRNA splicing, cell metabolism) [PMID: 20581084]:

> "its activity at promoter sites more often leads to the up-regulation of genes essential to protein biosynthesis and energy production" — Dejosez et al. 2010 [PMID: 20581084]

In zebrafish, loss of THAP11 (and *hcfc1b*) impairs proliferation/differentiation of neural precursors and neural crest, causing craniofacial cartilage loss that is **rescued by human *MMACHC*** [PMID: 25281006]:

> "the hcfc1b-mediated craniofacial abnormalities were rescued by expression of human MMACHC, a downstream target of HCFC1 that is aberrantly expressed in cblX" — Quintana et al. 2014 [PMID: 25281006]

This explains two key features of cblL: (1) **only hypomorphic alleles are compatible with life** (complete loss is lethal), and (2) the disorder is **syndromic/neurodevelopmental**, drawing on THAP11's broad transcriptional program in addition to the *MMACHC*-dependent cobalamin branch.

F006 — Epidemiology and inheritance: ultra-rare autosomal recessive, single reported homozygous case

Only **one patient** with cblL/MAHCL has been reported worldwide — a boy born to Moroccan parents carrying a homozygous *THAP11* c.240C>G (p.Phe80Leu) missense variant, implying **autosomal recessive inheritance with a likely consanguineous/founder background** [PMID: 28449119].

> "We report a patient who presented with clinical and biochemical phenotypic features that overlap cblX, but who does not have any mutations in either MMACHC or HCFC1." — Quintana et al. 2017 [PMID: 28449119]

No prevalence or incidence estimates exist; the disorder is at "unknown/ultra-rare" frequency. For contrast, the biochemically related **cblC (*MMACHC*) is the most common inborn error of B12 metabolism**, with a newborn-screening incidence for MMA estimated at ~1:3,920 live births in one Chinese cohort [PMID: 26563984].

F007 — Verified molecular coordinates and protein-domain context of the cblL variant

The single cblL-causing variant is **NM_020457.3:c.240C>G, p.(Phe80Leu)**, located at **GRCh38 chr16:67,842,794** (GRCh37 chr16:67,876,697), cytoband **16q22.1**; ClinVar Variation 393304 (VCV000393304), dbSNP **rs188675529**, OMIM 609119.0001, ClinGen CA396396088. The 1000 Genomes global minor-allele frequency is **~0.0032 (~0.3%)** — a low-frequency (not private) allele that is pathogenic only in the **biallelic** state. **Phe80** sits within the highly conserved **THAP-type zinc-finger DNA-binding domain** of THAP11; the protein also contains Gln-rich and Ala-rich regions, an HCFC1-binding motif (HBM), and a coiled-coil domain (Quintana et al. 2017 [PMID: 28449119]; ClinVar/UniProt Q96EK4).

Notably, of three cblX-like, *MMACHC*/*HCFC1*-negative subjects investigated, **only one** carried the THAP11 variant — underscoring genetic heterogeneity within the cblX-like group and the singular nature of the confirmed cblL case.

Section-by-Section Report

1. Disease Information

- **What it is:** An autosomal-recessive inborn error of intracellular cobalamin (vitamin B12) metabolism in which impaired transcriptional activation of *MMACHC* (by mutant THAP11) produces a combined methylmalonic aciduria + homocystinuria biochemical phenotype superimposed on a severe syndromic neurodevelopmental disorder.
- **Key identifiers:** OMIM #**620940** (MAHCL); MONDO **:0975798**; gene *THAP11* OMIM 609119 / HGNC:20856. No dedicated Orphanet, ICD-10/ICD-11, or MeSH code has been established for cblL specifically; it is generally grouped under disorders of cobalamin metabolism (pragmatically coded within the E71/E72 metabolic families — no cblL-specific code exists; *information not available*).
- **Synonyms/alternative names:** Methylmalonic aciduria and homocystinuria, cblL type; MAHCL; THAP11-related cobalamin metabolism disorder; a "cblX-like" cobalamin disorder.
- **Source of information: Aggregated disease-level resources** (OMIM, ClinVar, primary literature) plus a **single individual patient** report — not EHR/population data.

2. Etiology

- **Primary cause (genetic):** Biallelic hypomorphic **missense** variants in *THAP11* (the single confirmed allele is p.Phe80Leu in the THAP DNA-binding domain) [PMID: 28449119]. The mechanism is loss of transcriptional activation of *MMACHC* (F002). There is no environmental or infectious cause; cobalamin deficiency here is *functional/intracellular*, not dietary.
- **Genetic risk factors:** Homozygosity/compound heterozygosity for pathogenic *THAP11* variants; **consanguinity/founder background** (Moroccan parentage in the index case) increases the chance of homozygosity for the low-frequency allele (MAF ~0.3%).
- **Environmental risk / protective factors:** None established for cblL specifically. By analogy to cblC, adequate parenteral cobalamin and methyl-donor supplementation is protective against metabolic decompensation; catabolic stressors (infection, fasting) may precipitate crises (*inferred from cblC*).
- **Gene–environment interactions:** Not characterized. Plausibly, cobalamin/betaine/folate status modulates biochemical severity (*inferred*).

3. Phenotypes

Because only one patient has been reported, phenotype frequencies for cblL itself are anecdotal (n=1); the table below marks cblL-observed features and cblC-extrapolated features.

Phenotype	Type	Onset	Severity	Source	Suggested HPO
Myoclonic seizures	Clinical sign	Neonatal/ infantile (2 mo)	Severe	cblL (n=1) [PMID:28449119]	HP:0002123 (myoclonic seizure
Global developmental impairment / delay	Symptom	Infantile	Severe	cblL (n=1) [PMID:28449119]	HP:0001263
Methylmalonic aciduria	Lab abnormality	Infantile	Mild	cblL (n=1) [PMID:28449119]	HP:0012120
Homocystinuria / hyperhomocysteinemia	Lab abnormality	—	Not documented in index case	cblC-extrapolated	HP:0002160
Low-normal plasma methionine	Lab abnormality	Infantile	—	cblL (n=1) [PMID:28449119]	HP:0500152 (hypomethioninem
Craniofacial abnormalities (model)	Physical	Embryonic (model)	—	zebrafish [PMID:25281006]	HP:0001999
Intellectual disability	Symptom	Childhood	Moderate–severe (cblC)	cblC/cblX-extrapolated	HP:0001249
Feeding difficulty, hypotonia, lethargy	Symptom/ sign	Infantile	Variable (cblC)	cblC-extrapolated [PMID:20924684]	HP:0011968, HP:0001252, HP:0001254
Megaloblastic anemia	Lab abnormality	Variable (cblC)	Variable	cblC-extrapolated	HP:0001889

- **Progression:** Presumed progressive/episodic with metabolic decompensations, as in cblC.
- **Quality-of-life impact:** Severe — profound developmental impairment and seizures imply high dependency; formal QoL instruments have not been applied to cblL (*information not available*).

4. Genetic/Molecular Information

- **Causal gene:** ***THAP11*** (16q22.1; OMIM 609119; HGNC:20856; UniProt **Q96EK4**).
- **Pathogenic variant (only one confirmed):** NM_020457.3:**c.240C>G**, **p.(Phe80Leu)**; GRCh38 chr16:67,842,794; ClinVar VCV000393304; dbSNP rs188675529; OMIM 609119.0001. **Type:** missense. **Classification:** pathogenic in the biallelic state (per the disease-defining report; ClinVar entry present). **Allele frequency:** ~0.0032 (1000 Genomes global) — low-frequency, not private. **Origin:** germline. **Functional consequence:** hypomorphic loss of function affecting the THAP zinc-finger DNA-binding domain → reduced transcriptional activation of *MMACHC* (F002, F007).
- **Modifier genes:** *HCFC1* and *ZNF143* are obligate partners in the same transcriptional complex; variation in these (or in *MMACHC* itself) could plausibly modify severity (*inferred*). No formal modifier data for cblL.
- **Epigenetic information:** Not established for cblL. (In the related "epi-cblC," aberrant *PRDX1* transcripts methylate the *MMACHC* promoter [PMID: 42002808] — a parallel example of *MMACHC* silencing by a non-coding mechanism, illustrating that *MMACHC* output can be reduced without coding mutations.)
- **Chromosomal abnormalities:** None; cblL is a single-nucleotide/point-variant disorder.

5. Environmental Information

No environmental, lifestyle, or infectious agents cause cblL. As with other combined MMA/HC disorders, catabolic triggers (intercurrent infection, fasting, high protein load) may precipitate metabolic decompensation (*inferred from cblC* [PMID: 20924684]). Not applicable: toxins, radiation, occupational exposure, pathogens.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic hypomorphic **THAP11 p.Phe80Leu** variants impair the THAP zinc-finger DNA-binding domain → **reduced THAP11 transcription-factor function** (demonstrated: variant + fibroblast RNA-seq) [PMID: 28449119].
2. Impaired THAP11 **weakens the THAP11–HCFC1(–ZNF143) transcriptional complex** at target promoters → **decreased transcriptional activation of *MMACHC*** (demonstrated in cblL fibroblasts; established in cblX) [PMID: 28449119; 24011988].
3. Lower *MMACHC* mRNA/protein → **loss of cytosolic cobalamin chaperone activity** (reductive decyanation + glutathione-dependent dealkylation) [PMID: 19801555].

4. Failure to process cobalamin → **deficiency of both active cofactors**: methylcobalamin and adenosylcobalamin (inferred from the biochemistry of the pathway). 5a. AdoCbl deficiency → **methylmalonyl-CoA mutase (MMUT) dysfunction** → **methylmalonic acid accumulation** → methylmalonic aciduria [PMID: 28449119]. 5b. MeCbl deficiency → **methionine synthase (MTR) dysfunction** → **homocysteine accumulation** + **low methionine** (homocystinuria/hypomethioninemia; expected, mildly expressed in index case) [PMID: 19801555].
5. **Branch (MMACHC-independent):** reduced THAP11 also **dysregulates its broad program of protein-biosynthesis, energy-metabolism, and neural/neural-crest developmental genes** → contributes to the **severe syndromic neurodevelopmental phenotype** independent of the cobalamin defect (demonstrated in mouse/zebrafish; inferred for the human syndrome) [PMID: 18585351; 20581084; 25281006].
6. Convergence of metabolite toxicity (MMA, homocysteine) + impaired methylation (low methionine/SAM) + developmental gene dysregulation → **infantile myoclonic seizures and profound global developmental impairment** [PMID: 28449119].

Molecular pathways: cobalamin (B12) intracellular processing pathway; methionine/homocysteine remethylation cycle; methylmalonate–succinate (propionate catabolism) pathway; THAP11/HCF-1 transcriptional enhancer program (Reactome: Metabolism of vitamins and cofactors; cobalamin metabolism). **Cellular processes:** transcriptional regulation, cell proliferation/differentiation (neural precursors, neural crest), cellular cobalamin trafficking. **Protein dysfunction:** hypomorphic loss of THAP11 DNA-binding function (missense in THAP domain); downstream loss of MMACHC chaperone function. **Metabolic changes:** amino-acid (methionine/homocysteine) and organic-acid (methylmalonate) metabolism; impaired one-carbon/methylation metabolism. **Immune involvement:** none primary. **Tissue-damage mechanisms:** presumed metabolite-mediated neurotoxicity and impaired neurodevelopment (*inferred*).

Suggested ontology terms: GO:0006355 (regulation of transcription), GO:0009235 (cobalamin metabolic process), GO:0006555 (methionine metabolic process), GO:0032259 (methylation); CL:0000047 (neuronal stem cell), CL:0000333 (migratory neural crest cell); CHEBI:16009 (methylmalonic acid), CHEBI:17230 (homocysteine).

7. Anatomical Structures Affected

- **Organ level (primary):** brain/central nervous system (UBERON:0000955 brain; UBERON:0001017 central nervous system) — seizures, developmental impairment. **Body system:** nervous system (UBERON:0001016).

- **Secondary/model-based:** craniofacial cartilage/skeleton (Meckel's cartilage in zebrafish; UBERON:0004744) — neural-crest-derived structures.
- **Tissue/cell level:** neural precursor cells and neural crest cells (CL:0000047; CL:0000333); by analogy to cblC, hematopoietic (megaloblastic changes) and retinal cells may be affected (*inferred*).
- **Subcellular level: cytosol** (GO:0005829 — site of MMACHC chaperone activity) and **nucleus** (GO:0005634 — site of THAP11 transcriptional activity). Downstream mitochondrial enzyme (MMUT) function is secondarily impaired (GO:0005739 mitochondrion).
- **Localization/laterality:** CNS involvement is bilateral/diffuse (*inferred*).

8. Temporal Development

- **Onset: Infantile/neonatal** — the index patient presented at **2 months** with seizures [PMID: 28449119]. Onset pattern: subacute–progressive.
- **Progression:** presumed **progressive** with superimposed **episodic** metabolic decompensations, as in cblC; disease duration is **chronic/lifelong**.
- **Stages:** not formally staged.
- **Critical periods:** early infancy is the key window for intervention; by analogy to cblC, earlier treatment initiation (ideally via newborn screening) is associated with better — though still frequently impaired — neurodevelopmental outcome [PMID: 26563984].

9. Inheritance and Population

- **Inheritance: Autosomal recessive** (homozygous index case) [PMID: 28449119].
- **Epidemiology: Ultra-rare — a single reported case worldwide.** No prevalence/incidence figures exist. Contrast with cblC (most common B12 inborn error; MMA newborn-screening incidence ~1:3,920 in one Chinese cohort) [PMID: 26563984].
- **Penetrance/expressivity:** presumed complete penetrance in biallelic state; expressivity unknown (n=1).
- **Founder effect/consanguinity:** likely — Moroccan parentage and homozygosity for a low-frequency (~0.3%) allele suggest a consanguineous/founder background [PMID: 28449119; F007].
- **Carrier frequency:** the p.Phe80Leu allele has a 1000 Genomes global MAF ~0.0032; carrier frequency for *cblL as a disease* is unknown given only one causal allele is documented.

- **Demographics:** single male patient of North African (Moroccan) descent; no sex-ratio or age-distribution data can be derived from n=1.

10. Diagnostics

- **Biochemical/laboratory:** elevated **methylmalonic acid** (blood/urine); **elevated total homocysteine** with **low/low-normal methionine**; elevated **propionylcarnitine (C3)** and ratios on acylcarnitine profiling (MS/MS) — the classic combined MMA+HC signature [PMID: 20924684; 28449119]. In the index cblL case, MMA was mild and methionine low-normal.
- **Complementation/cellular studies:** cblL shows a **cblC-like biochemical complementation pattern** with reduced *MMACHC* expression on fibroblast RNA-seq despite a normal *MMACHC* sequence — a key clue that directs testing to the regulatory genes.
- **Genetic testing (definitive):** Because the biochemistry mimics cblC/cblX, diagnosis requires **sequencing beyond *MMACHC***. Recommended approach: a **combined MMA+HC gene panel or exome** covering *MMACHC, MMADHC, LMBRD1, ABCD4, HCFC1, ZNF143,* and ***THAP11*** [PMID: 34655177]. cblL is confirmed by finding **biallelic *THAP11*** variants after *MMACHC* and *HCFC1* are excluded [PMID: 28449119]. WES/WGS and targeted panels are all appropriate; single-gene *THAP11* testing is reserved for the cblX-like, *MMACHC*/*HCFC1*-negative phenotype.
- **Imaging/electrophysiology:** EEG for seizure characterization; brain MRI as per cblC workup (white-matter/structural changes) — not specifically reported for cblL (*information not available*).
- **Differential diagnosis:** cblC (*MMACHC*), cblX (*HCFC1*), cblD/cblF/cblJ, ZNF143-related cblX-like disease, epi-cblC (*PRDX1* epimutation [PMID: 42002808]), and dietary/maternal B12 deficiency. The distinguishing feature of cblL is **normal *MMACHC*/*HCFC1* with biallelic *THAP11* variants and reduced *MMACHC* expression.**
- **Screening:** MMA/homocysteine-based **newborn screening** (C3, C3/C2, methionine, tHcy) will flag the biochemical phenotype but cannot distinguish cblL from other combined MMA+HC disorders without molecular confirmation [PMID: 26563984].

Suggested LOINC-type analytes: methylmalonic acid (urine/plasma), total homocysteine, methionine, propionylcarnitine (C3).

11. Outcome/Prognosis

- **Survival/mortality:** Unknown for cblL (n=1, no long-term follow-up reported). In cblC, mortality occurs from metabolic crises (e.g., a screened cblC infant died at 38 days from infection-triggered crisis) [PMID: 26563984].
- **Morbidity/function:** The index patient had **severe global developmental impairment and seizures**, predicting high long-term disability [PMID: 28449119]. By analogy to cblC, **cognitive and visual impairment are frequent despite treatment** [PMID: 23746552].
- **Disease course:** presumed chronic with risk of decompensation; recovery potential is limited given the severe neurodevelopmental component and THAP11's developmental role.
- **Prognostic factors:** earlier diagnosis/treatment and biochemical control are favorable in cblC but do not guarantee normal neurodevelopment [PMID: 26563984]. cblL-specific prognostic biomarkers are undefined.

12. Treatment

No cblL-specific trials exist; management is **rational extrapolation from cblC/cblX**:

Intervention	Rationale	Evidence	Suggested NCIT
Hydroxocobalamin (OHCbl), parenteral	Supplies cobalamin to partially overcome reduced processing; mainstay of cblC	[PMID: 23746552]	NCIT:C29094 (Hydroxocobalamin)
Betaine	Remethylates homocysteine → methionine (bypasses MTR limitation)	[PMID: 23746552]	NCIT:C61796 (Betaine)
Folinic acid	Supports one-carbon/ methylation metabolism	[PMID: 23746552]	NCIT:C61837 (Leucovorin/folinic acid)
Levocarnitine (L-carnitine)	Conjugates/clears propionyl/ methylmalonyl species	[PMID: 23746552]	NCIT:C61785 (Levocarnitine)
Dietary protein moderation	Limits propiogenic amino-acid load	[PMID: 23746552]	NCIT:C15222 (Dietary intervention)

- **Pharmacogenomics:** none established.

- **Advanced/experimental therapeutics:** Because cblL is a *transcriptional* deficiency of *MMACHC*, conceptually distinctive approaches (e.g., restoring *MMACHC* expression, gene/gene-regulation therapy) are of theoretical interest but **not developed**; no gene, cell, or RNA therapy exists for cblL. The zebrafish rescue by human *MMACHC* [PMID: 25281006] provides proof-of-concept that restoring the downstream target can correct at least the *MMACHC*-dependent (craniofacial) branch — but not necessarily the broader THAP11 program.
- **Treatment outcomes:** biochemical improvement is achievable (as in cblC), but neurodevelopmental outcomes are frequently poor [PMID: 23746552; 26563984]. Hydroxocobalamin dose escalation improved biochemistry in a cblC series [PMID: 23746552].

13. Prevention

- **Primary prevention:** not applicable (genetic disorder). **Genetic counseling** for at-risk (consanguineous/carrier) families and **carrier/cascade testing** for the familial *THAP11* variant are the principal preventive tools.
- **Secondary prevention: newborn screening** (MMA/homocysteine metabolites) enables early detection of the combined MMA+HC phenotype and early treatment initiation [PMID: 26563984]; molecular confirmation identifies cblL specifically.
- **Prenatal/preimplantation:** prenatal diagnosis by chorionic villus sampling / molecular testing is standard for known familial cobalamin-disorder variants (demonstrated for *MMACHC* [PMID: 30157807; 26149271]) and is applicable to a known familial *THAP11* variant.
- **Tertiary prevention:** metabolic-crisis prevention through consistent cofactor supplementation, sick-day protocols, and avoidance of catabolic stress (*inferred from cblC*).

14. Other Species / Natural Disease

- **Taxonomy of study organisms:** *Mus musculus* (NCBI:txid10090); *Danio rerio* (zebrafish; NCBI:txid7955).
- **Orthologous genes:** mouse *Thap11* (Ronin); zebrafish *thap11* and *hcfc1b* (paralog of human *HCFC1*) [PMID: 25281006].
- **Natural disease in other species:** No naturally occurring THAP11-cobalamin disorder has been reported in companion animals or wildlife (*information not available*; no OMIA entry known for cblL). Not zoonotic; not transmissible.

- **Comparative biology / conservation:** The THAP11–HCF-1 axis and its control of biosynthetic/energy genes are **evolutionarily conserved** from zebrafish to mammals; the zebrafish craniofacial phenotype is rescued by **human *MMACHC***, demonstrating conserved regulatory logic [PMID: 25281006; 20581084].

15. Model Organisms

Model	Type	Key phenotype	Recapitulation	Reference
Thap11/ Ronin-null mouse	Mammalian knockout	Peri-implantation lethality; inner-cell-mass defects	Demonstrates essentiality; **too severe** to model viable human hypomorphic disease	[PMID: 18585351]
Ronin/ HCF-1 ChIP + ESC studies	In vitro / mouse ESC	Binds hyperconserved enhancer; regulates protein-biosynthesis/ energy genes	Defines the broad transcriptional program disrupted in cblL	[PMID: 20581084]
Zebrafish *thap11* / *hcfc1b* loss	Vertebrate morphant/ mutant	Craniofacial cartilage loss (Meckel's cartilage), neural precursor/ neural-crest defects; **rescued by human *MMACHC***	Models the MMACHC-dependent developmental branch; links THAP11→MMACHC in vivo	[PMID: 25281006; 28449119]
cblL patient fibroblasts	In vitro (human)	Reduced *MMACHC* (and *TMOD2*) expression; cblC-like complementation	Directly demonstrates the disease mechanism	[PMID: 28449119]

- **Model limitations:** the mouse null is embryonic-lethal (cannot model postnatal neurodevelopmental disease); zebrafish captures the developmental/MMACHC-dependent branch but not the full human syndrome; no viable hypomorphic mammalian *Thap11* p.Phe80Leu knock-in has been reported — a clear resource gap.

Mechanistic Model / Interpretation

```
 THAP11 (p.Phe80Leu, biallelic)  ── impairs ──►  THAP zinc-finger DNA binding
              │
              ▼
 Weakened THAP11–HCFC1(–ZNF143) transcriptional complex
              │  (↓ activation at MMACHC promoter)
              ▼
 ↓ MMACHC mRNA/protein  ──(normal MMACHC gene sequence!)──►  regulatory phenocopy of cblC
              │
              ▼
 ↓ Cytosolic cobalamin chaperone activity (decyanation/dealkylation, Kim 2009)
              │
      ┌───────┴─────────┐
      ▼                 ▼
↓ AdoCbl            ↓ MeCbl
↓ MMUT activity     ↓ MTR (methionine synthase)
↑ Methylmalonic     ↑ Homocysteine
  acid (aciduria)   ↓ Methionine
      └───────┬─────────┘
              ▼
 Metabolite toxicity + impaired methylation
              │
              │   ┌── PARALLEL, MMACHC-INDEPENDENT BRANCH ──────────┐
              │   │ ↓ THAP11 program for protein biosynthesis, energy, │
              │   │ neural/neural-crest development (Dejosez 2008/2010, │
              │   │ Quintana 2014 zebrafish)                            │
              │   └─────────────────────────────────────────────────────┘
              ▼                          ▼
 Infantile myoclonic seizures + severe global developmental impairment
```

The unifying insight is that **cblL is not a defect of the cobalamin machinery itself but of its transcriptional supply line.** THAP11 sits *upstream* of *MMACHC*; the downstream biochemistry (MMA, homocysteine, methionine) is *identical in kind* to cblC but arises from **gene-expression failure rather than protein-coding mutation**. This dual nature — a metabolic branch (shared with cblC) plus a developmental/transcriptional branch (unique to loss of an essential pluripotency/biosynthesis factor) — predicts a phenotype that is **more syndromic and neurodevelopmentally severe** than metabolite toxicity alone would explain, consistent with the single reported case.

Evidence Base

PMID	Title (abbrev.)	Role in this report
28449119	*Mutations in THAP11 cause an inborn error of cobalamin metabolism and developmental abnormalities*	**Disease-defining paper**; identifies THAP11 c.240C>G/p.Phe80Leu; THAP11–HCFC1–MMACHC axis; zebrafish craniofacial data (F001, F002, F004, F006, F007)
24011988	*An X-linked cobalamin disorder caused by mutations in HCFC1*	Establishes that disrupting the HCFC1/THAP11 complex reduces *MMACHC* mRNA/protein (F002)
19801555	*A human vitamin B12 trafficking protein uses glutathione transferase activity...*	Defines MMACHC's decyanation/dealkylation chaperone role (F003)
18585351	*Ronin is essential for embryogenesis and pluripotency of mouse ESCs*	THAP11/Ronin essentiality; peri-implantation lethality; HCF-1 binding (F005)
20581084	*Ronin/Hcf-1 binds a hyperconserved enhancer...*	THAP11's broad biosynthesis/energy transcriptional program (F005)
25281006	*Hcfc1b regulates craniofacial development by modulating mmachc*	Zebrafish model; craniofacial phenotype rescued by human *MMACHC* (F005)
23746552	*Clinical/biochemical outcome after hydroxocobalamin dose escalation in cblC*	Treatment framework (OHCbl, betaine, folinic acid, carnitine) (F004)
34655177	*Prenatal diagnosis of cblC using clinical exome/targeted analysis*	Places THAP11 in the combined MMA+HC gene panel (F001)
26563984	*cblC in Shandong: incidence and outcomes*	cblC incidence (~1:3,920) and post-treatment neurodevelopmental outcomes (F006)
20924684	*Clinical/biochemical/molecular analysis of cblC in China*	Clinical/biochemical signature of combined MMA+HC (phenotype/diagnostics background)
30157807 / 26149271	*Prenatal genetic diagnosis of cblC*	Prenatal-diagnosis methodology applicable to familial variants (prevention)
42002808	*Epi-cblC with MMACHC promoter methylation (PRDX1)*	

PMID	Title (abbrev.)	Role in this report
		Parallel example of *MMACHC* silencing without coding mutation (differential/epigenetics)
33517344	*HCFC1 exon-skipping variant: ID without metabolic abnormalities*	Illustrates dissociation of clinical vs biochemical phenotype in the same regulatory pathway
35337626	*Inherited defects of cobalamin metabolism* (review)	Broad classification context for the cbl complementation groups
20631720	*Mutation spectrum of MMACHC in Chinese patients*	Founder-effect and mutation-spectrum context for the biochemically related cblC
23580368	*Novel deletion in late-onset cblC*	Genotype–phenotype/structure context for MMACHC

Evidence-type note: cblL-specific evidence is **human (single clinical case + patient fibroblasts)** plus **model organism (mouse, zebrafish)** and **in vitro biochemistry**. All clinical, prognostic, and therapeutic specifics beyond the index case are **extrapolated** from cblC/cblX human cohorts.

Limitations and Knowledge Gaps

1. **n = 1.** The entire cblL clinical phenotype rests on a single homozygous p.Phe80Leu patient. Frequencies, penetrance, expressivity, natural history, sex distribution, and full phenotype spectrum are essentially unknown.
2. **No population/epidemiology data.** No prevalence or incidence estimates; carrier frequency for cblL as a disease cannot be computed from one allele.
3. **Homocystinuria under-characterized in the index case.** Overt homocystinuria was not documented, leaving the "combined" designation partly inferred from the pathway.
4. **Treatment evidence is entirely extrapolated.** No cblL-specific treatment or outcome data exist; the response of a *transcriptional* defect to cobalamin supplementation may differ from cblC.

5. **No viable mammalian disease model.** The mouse null is embryonic-lethal; a hypomorphic knock-in modeling p.Phe80Leu has not been reported.
6. **Genetic heterogeneity within "cblX-like."** *ZNF143* and other regulators may cause overlapping disease; two of three cblX-like probands in the defining study lacked the THAP11 variant, indicating additional, still-uncharacterized genes.
7. **Ontology/identifier gaps.** No dedicated Orphanet/ICD/MeSH code exists for cblL specifically.

Proposed Follow-up Experiments / Actions

1. **Case ascertainment:** Deposit and query cblL/*THAP11* in GeneMatcher, ClinVar, and metabolic-disease registries to find additional biallelic *THAP11* patients; report their full biochemical (including tHcy) and neuroimaging phenotypes.
2. **Functional validation of variants:** Luciferase/ChIP and RNA-seq assays measuring *MMACHC* activation by WT vs p.Phe80Leu (and any new alleles) THAP11 in the THAP11–HCFC1–ZNF143 complex; quantify residual activity (hypomorph vs null).
3. **Hypomorphic mouse knock-in** of the p.Phe80Leu equivalent to establish a viable, treatable model that captures the neurodevelopmental phenotype (the null being lethal).
4. **Separate the two branches:** In zebrafish/organoids, test whether restoring *MMACHC* alone rescues metabolic vs developmental phenotypes, quantifying the MMACHC-independent contribution of THAP11's biosynthetic program.
5. **Therapeutic trials in model systems:** Compare hydroxocobalamin/betaine/folinic-acid/carnitine responses; explore approaches to upregulate residual *MMACHC* transcription (e.g., targeting the regulatory complex) as a cblL-specific strategy.
6. **Biomarker natural history:** Serial MMA, tHcy, methionine, and C3 in any identified cases to define treatment targets and prognostic thresholds.
7. **Ontology curation:** Establish dedicated Orphanet/ICD-11 mappings and confirm the MONDO/HPO annotation set for cblL in the knowledge base.

Report compiled from 7 confirmed findings and 17 reviewed papers over 5 investigation iterations. cblL-specific claims are grounded in Quintana et al. 2017 [PMID: 28449119] and supporting model/biochemical studies; all clinical, prognostic, and therapeutic specifics beyond the single index case are explicitly extrapolated from the biochemically equivalent cblC/cblX disorders and labeled as such.

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